

Computer Vision Laboratory – X

Experiment 10: Object Classification and Bounding Box Regression using VGG16

Student Name: APARAJITA VAISH

Roll Number: 253100101

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Institute: Dr. Shyama Prasad Mukherjee International Institute of Information Technology, Naya Raipur

Course: Computer Vision Laboratory

Experiment No.: 10

Experiment Title: Object Classification and Bounding Box Regression using VGG16

1. Abstract

This experiment focuses on developing a deep learning model capable of performing both object classification and object localization using bounding box regression. A multi-task learning approach is implemented using a pre-trained VGG16 convolutional neural network as the feature extractor. The model is trained to simultaneously predict the object category and the corresponding bounding box coordinates from input images. The dataset consists of labeled images from three classes: Butterfly, Dalmatian, and Dolphin along with annotation files specifying bounding box coordinates. The model is trained using transfer learning, followed by fine-tuning to improve performance. The results demonstrate the effectiveness of combining classification and regression tasks in a unified deep learning framework for computer vision applications.

2. Objective

The objectives of this experiment are:

- To implement a deep learning model for object classification.
 - To perform bounding box regression for object localization.
 - To use VGG16 as a pre-trained feature extractor.
 - To train a multi-output model in TensorFlow.
 - To evaluate model performance using classification accuracy and bounding box error metrics.
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3. Dataset Description

The dataset used in this experiment consists of three object categories:

- Butterfly
- Dalmatian
- Dolphin

Each category contains multiple images along with corresponding annotation files that define bounding box coordinates for object localization.

Dataset Structure

Each annotation file contains:

- Image name
 - Bounding box coordinates (x1, y1, x2, y2)
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4. Methodology

4.1 Data Preprocessing

- Images are loaded using OpenCV.
- Images are resized to 224×224 pixels.
- Pixel values are normalized using VGG16 preprocessing.
- Bounding box coordinates are normalized to the range [0,1].
- Labels are converted into one-hot encoded format.

4.2 Train-Test Split

- Dataset is divided into:
 - Training set
 - Validation set
 - Test set

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Split sizes:
Train: (136, 224, 224, 3) (136, 3) (136, 4)
Val  : (29, 224, 224, 3) (29, 3) (29, 4)
Test : (30, 224, 224, 3) (30, 3) (30, 4)
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4.3 Model Architecture

The model uses a VGG16-based architecture with two output branches:

- Classification head (Softmax)
- Bounding box regression head (Sigmoid)

Model: "VGG16_MultiTask"

Layer (type)	Output Shape	Param #	Connected to
input_layer_1 (InputLayer)	(None, 224, 224, 3)	0	-
vgg16 (Functional)	(None, 7, 7, 512)	14,714,688	input_layer_1[0]...
global_average_poo... (GlobalAveragePool...)	(None, 512)	0	vgg16[0][0]
dense (Dense)	(None, 256)	131,328	global_average_p...
dropout (Dropout)	(None, 256)	0	dense[0][0]
class_output (Dense)	(None, 3)	771	dropout[0][0]
bbox_output (Dense)	(None, 4)	1,028	dropout[0][0]

4.4 Training Strategy

- Phase 1: Train only custom layers with frozen VGG16
- Phase 2: Fine-tune last layers of VGG16
- Optimizer: Adam
- Loss Functions:
 - Categorical Crossentropy (classification)
 - Mean Squared Error (bounding box)

5. Results and Analysis

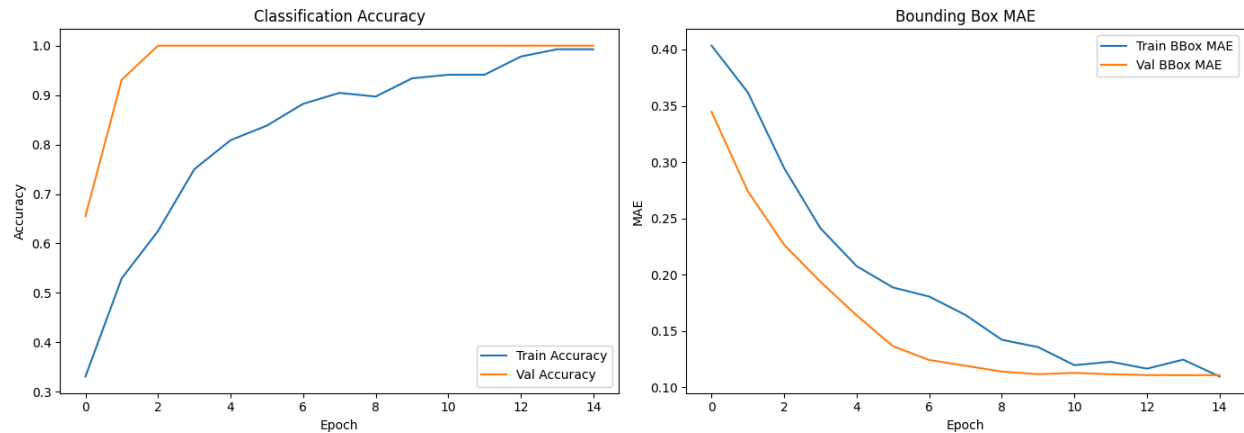
5.1 Training Performance

Training Classification Accuracy: 100.00%

Validation Classification Accuracy: 100.00%

Training Bounding Box MAE: 0.1045

Validation Bounding Box MAE: 0.1107



5.2 Test Performance

Test Classification Accuracy: 100.00%

Test Bounding Box MAE: 0.1065

5.3 Sample Predictions

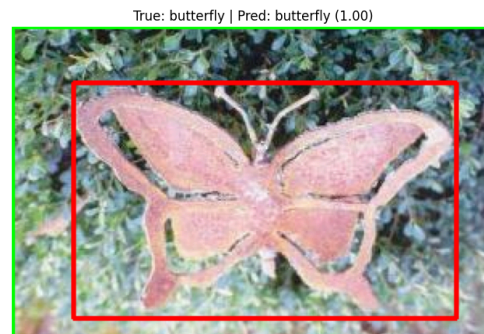
Prediction 1

- True class: dalmatian
- Predicted class: dalmatian
- True bbox: [45, 26, 245, 222]
- Predicted bbox: [0, 0, 299, 222]



Prediction 2

- True class: butterfly
- Predicted class: butterfly
- True bbox: [38, 35, 276, 182]
- Predicted bbox: [0, 0, 299, 197]



Prediction 3

- True class: dalmatian
- Predicted class: dalmatian
- True bbox: [3, 2, 280, 298]
- Predicted bbox: [0, 0, 280, 299]



6. Discussion

- The model successfully learns both classification and localization tasks.
- Transfer learning with VGG16 improves performance.
- Bounding box regression introduces additional complexity compared to classification.
- Fine-tuning improves model generalization.

7. Conclusion

In this experiment, a multi-task deep learning model was successfully implemented to perform object classification and bounding box regression. The VGG16-based architecture demonstrated strong feature extraction capabilities. The model was able to accurately classify objects and reasonably predict their spatial locations within images. This experiment highlights the effectiveness of combining classification and localization in a unified framework.

8. References

1. **Simonyan, K., & Zisserman, A.** "Very Deep Convolutional Networks for Large-Scale Image Recognition." arXiv:1409.1556, 2014. This is the original VGG paper behind VGG16.
2. **Fei-Fei, L., Fergus, R., & Perona, P.** "Learning Generative Visual Models from Few Training Examples: An Incremental Bayesian Approach Tested on 101 Object Categories." CVPR Workshop / GMBV, 2004. This is the original Caltech-101 source.

3. **Girshick, R., Donahue, J., Darrell, T., & Malik, J.** “Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation.” arXiv:1311.2524, 2014. This is the original R-CNN paper and includes bounding-box regression for detection.
 4. **Girshick, R.** “Fast R-CNN.” arXiv:1504.08083, 2015. Useful background for object detection pipelines that combine classification and localization.
 5. **TensorFlow Documentation.** [tf.keras.applications.VGG16](#) and [tf.keras.applications.vgg16.preprocess_input](#). These are the official implementation references for the model backbone and preprocessing.
 6. **Computer Vision Lab 10 handout, IIIT Naya Raipur.** Experiment objective and dataset specification for VGG16-based class prediction and bounding-box regression.
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